CS23532-COMPUTER NETWORKS-LAB MANUAL

Practical -4

AIM: Setup and configure a LAN (Local area network) using a Switch and Ethernet cables in your lab.

Experiment No. 4

Experiment Name

Setup and Configure a LAN Using Switch and Ethernet Cables

Aim

To set up and configure a Local Area Network (LAN) by connecting computers through a switch and Ethernet cables, assign IP addresses, test connectivity, and share files and folders between computers.

Procedure

1. Connect three or four computers to a network switch using Ethernet cables.
2. Ensure that all Ethernet cables are properly connected and the link indicators on the switch and computers are active.
3. Configure a unique IP address for each computer in the same network. For example:
    - PC1: 192.168.1.2
    - PC2: 192.168.1.3
    - PC3: 192.168.1.4
    - PC4: 192.168.1.5

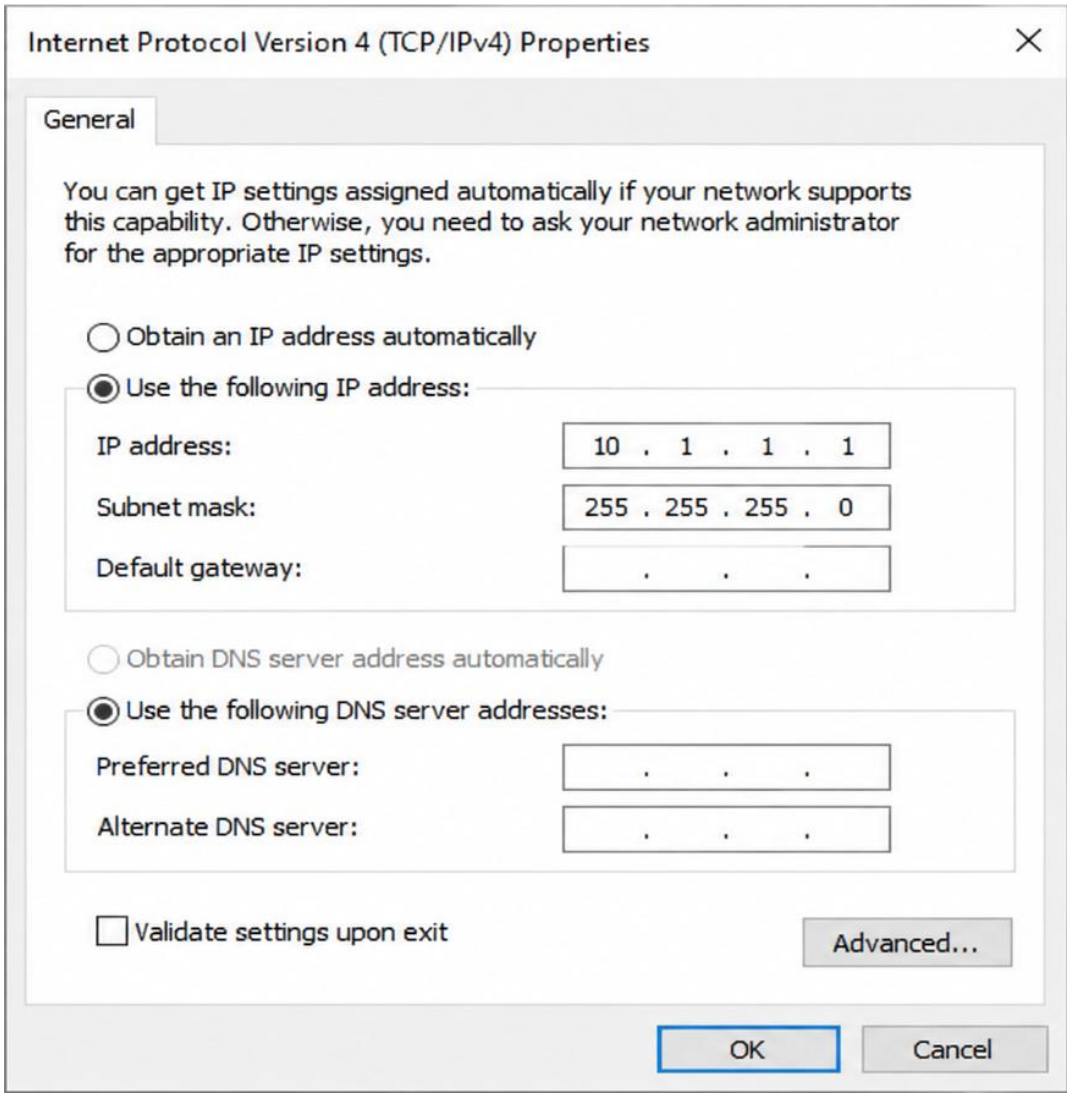
Set the subnet mask to 255.255.255.0 on all computers.

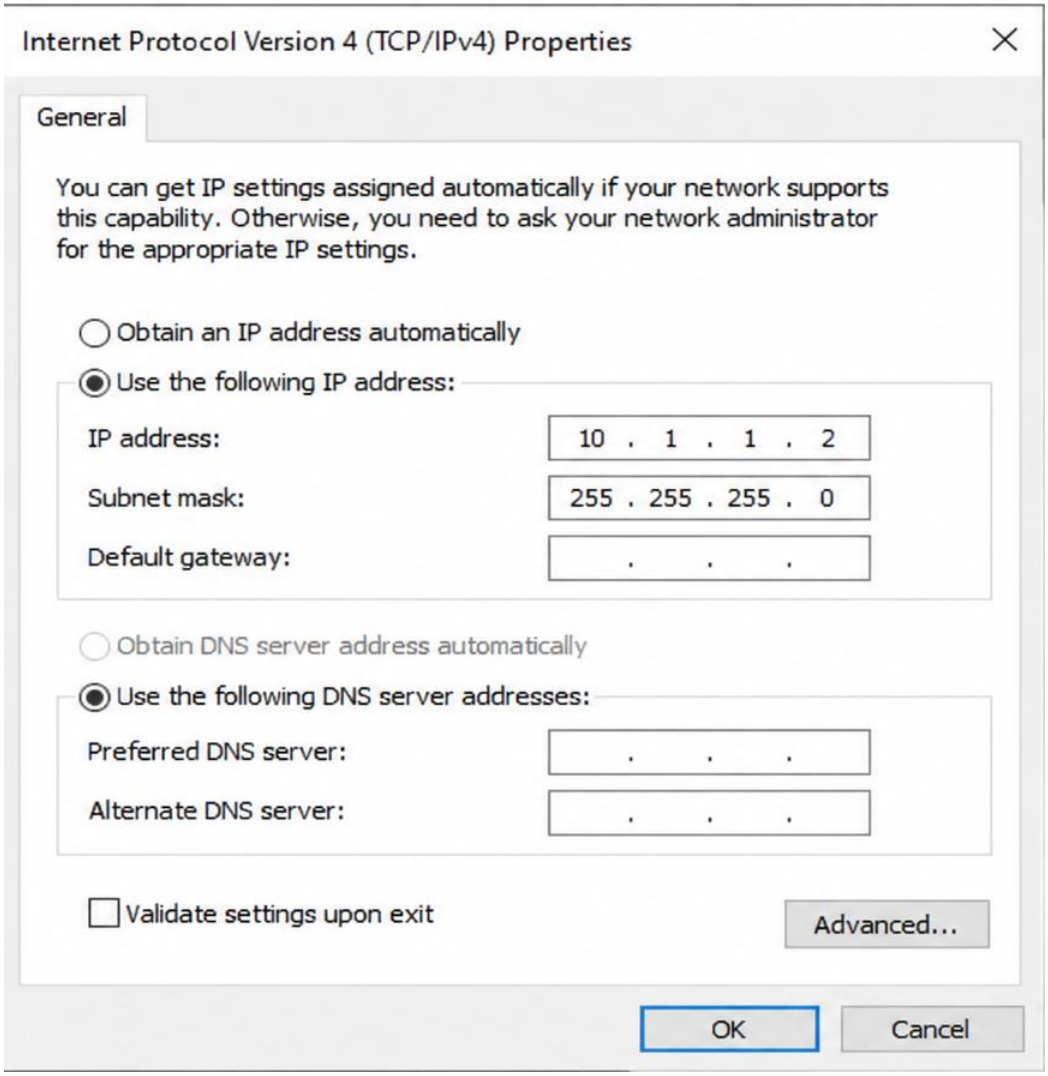
4. Open the Command Prompt on each computer and use the ipconfig command to verify the assigned IP address.
5. Test connectivity between computers using the ping command. For example, from PC1, execute:

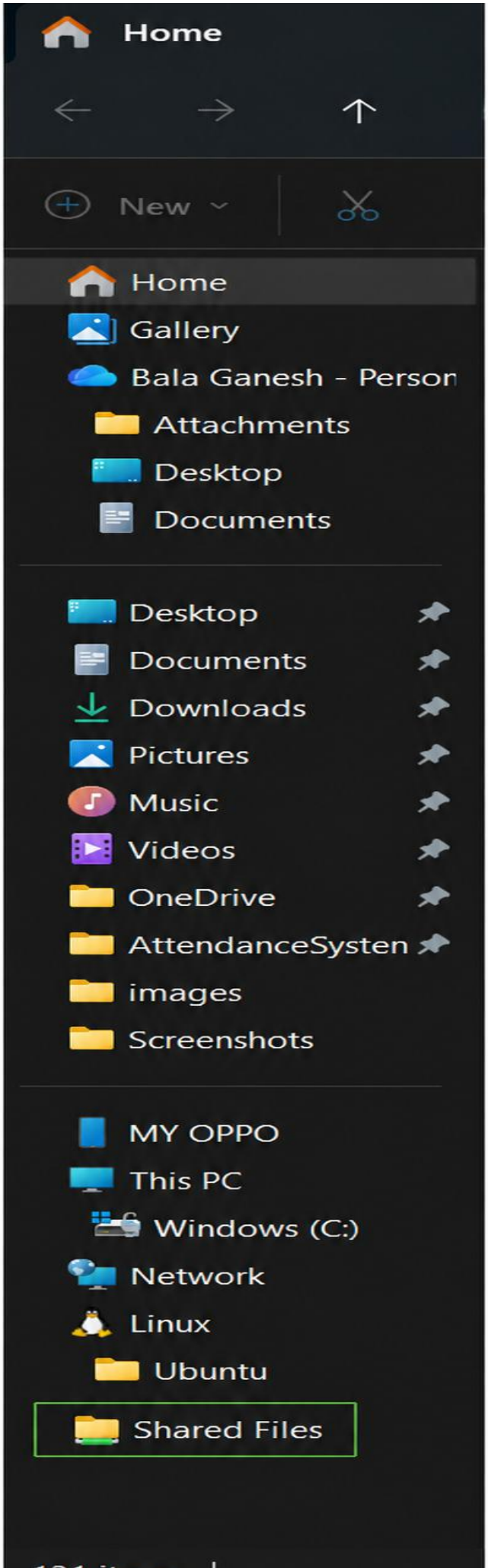
ping 192.168.1.3

Repeat the test for the other computers and verify that replies are received.

6. If the ping fails, check the IP addresses, subnet masks, Ethernet connections, and firewall settings.
7. Enable network discovery and file and printer sharing in the operating system's network settings.
8. Create a test folder on one computer and share it with the required network users. Access the shared folder from another computer using its network name or IP address.
9. Verify that the shared folder can be accessed successfully and record the observations.




Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address:
10 . 1 . 1 . 2
Subnet mask:
255 . 255 . 255 . 0
Default gateway:
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Validate settings upon exit
Advanced...
OK
Cancel


Home
New
Home
Gallery
Bala Ganesh - Person
Attachments
Desktop
Documents
Desktop
Documents
Downloads
Pictures
Music
Videos
OneDrive
AttendanceSysten
images
Screenshots
MY OPPO
This PC
Windows (C:)
Network
Linux
Ubuntu
Shared Files

Result

A Local Area Network was successfully configured by connecting computers through a switch and Ethernet cables. IP addresses were assigned, connectivity was tested using the ping command, and file and folder sharing between computers was verified.